

# markdown2html5-base

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## Full Markdown Functionality Reference

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This reference covers all `markdown2html5-base` features.

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### Programmatic Usage

---

```
from markdown2html5_base import MarkdownToHTML
converter = MarkdownToHTML()
html = converter.convert("# Hello")
print(html)
```

### 1. Headings (H1-H6)

---

| Markdown        | Output HTML        |
|-----------------|--------------------|
| # Heading 1     | <h1>Heading 1</h1> |
| ## Heading 2    | <h2>Heading 2</h2> |
| ### Heading 3   | <h3>Heading 3</h3> |
| #### Heading 4  | <h4>Heading 4</h4> |
| ##### Heading 5 | <h5>Heading 5</h5> |
| ##### Heading 6 | <h6>Heading 6</h6> |

Custom ID: ## Section {#sec1} => <h2 id="sec1">Section</h2>

## 2. Inline Text Styling

| Markdown          | Output HTML                           |
|-------------------|---------------------------------------|
| ***bold italic*** | <strong><em>bold italic</em></strong> |
| ___bold italic___ | <strong><em>bold italic</em></strong> |
| **bold**          | <strong>bold</strong>                 |
| __bold__          | <strong>bold</strong>                 |
| *italic*          | <em>italic</em>                       |
| _italic_          | <em>italic</em>                       |
| ~~strikethrough~~ | <s>strikethrough</s>                  |
| ==highlight==     | <mark>highlight</mark>                |
| ~subscript~       | <sub>subscript</sub>                  |
| ^^underline^^     | <u>underline</u>                      |
| ^superscript^     | <sup>superscript</sup>                |
| `code`            | <code>code</code>                     |

## 3. Links and Images

| Markdown               | Output HTML                                                                                |
|------------------------|--------------------------------------------------------------------------------------------|
| [text](url)            | <a href="url">text</a>                                                                     |
| ![alt](src)            | <figure></figure>                                                 |
| ![alt](src<br>"Title") | <figure><br><figcaption>Title</figcaption></figure> |

## 4. Fenced Code Blocks

A language tag after the opening fence (three backticks before and after the code) is rendered as a `<div class="code-lang">&sol;python&sol;</div>` label above the `<code>` element:

```
<div class="code-lang">&sol;python&sol;</div><pre><code>def hello():  
    print("Hi")</code></pre>
```

Without a language tag, the code renders plainly as `<pre><code>`.

## 5. Horizontal Rules

--- / \*\*\* / \_\_\_ (3+ chars) => `<hr>`

## 6. Lists

1. **Unordered:** \* Item or - Item => `<ul><li>Item</li></ul>`

2. **Ordered:** 1. Item => `<ol><li>Item</li></ol>`

3. **Task:**

\* [ ] todo => `<li><input type="checkbox" disabled> todo</li>`

\* [x] done => `<li><input type="checkbox" checked disabled> done</li>`

## 7. Blockquotes

> text => `<blockquote><p>text</p></blockquote>` : blank lines within a blockquote split into separate `<p>` tags.

## 8. Tables

Supports `<thead>`, `<tbody>`, `<tfoot>`, and alignment:

| Separator | Alignment |
|-----------|-----------|
| :---      | left      |
| :---:     | center    |
| ---:      | right     |

Footer: a row of = signs below the separator columns, after the body rows, renders a `<tfoot>`.

|          |       |        |  |
|----------|-------|--------|--|
| Product  | Qty   | Price  |  |
| :-----   | :-:   | ----:  |  |
| Apples   | 2     | \$3.00 |  |
| Bananas  | 3     | \$1.50 |  |
| Cherries | 1     | \$4.00 |  |
| =====    | ===== | =====  |  |
| Total    | 6     | \$8.50 |  |

=>

```

<table>
  <thead>
    <tr>
      <th style="text-align:left;">Product</th>
      <th style="text-align:center;">Qty</th>
      <th style="text-align:right;">Price</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td style="text-align:left;">Apples</td>
      <td style="text-align:center;">2</td>
      <td style="text-align:right;">$3.00</td>
    </tr>
    <tr>
      <td style="text-align:left;">Bananas</td>
      <td style="text-align:center;">3</td>
      <td style="text-align:right;">$1.50</td>
    </tr>
    <tr>
      <td style="text-align:left;">Cherries</td>
      <td style="text-align:center;">1</td>
      <td style="text-align:right;">$4.00</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td style="text-align:left;">Total</td>
      <td style="text-align:center;">6</td>
      <td style="text-align:right;">$8.50</td>
    </tr>
  </tfoot>
</table>

```

## 9. Definition Lists

---

Term  
: Definition 1  
: Definition 2

=>

```
<dl>
  <dt>Term</dt>
  <dd>Definition 1</dd>
  <dd>Definition 2</dd>
</dl>
```

## 10. Footnotes

Reference: `[^1]` => `<sup id="fnref:1"><a href="#fn:1" class="footnote-ref">1</a></sup>`

Definition: `[^1]: Text` at bottom => rendered in `<div class="footnotes"><ol>...</ol></div>`

## 11. Language Markers

Annotate blocks or inline text with a language using a valid BCP 47 tag (e.g. `de`, `fr`, `zh-Hans`).

- **Block level:** Place `{:lang}` (with a space after it) at the very start of a line, before the Markdown tag. It renders as the global `lang` attribute on the resulting element and needs no closing marker:

```
{:de} # Überschrift
{:fr} Un paragraphe français.
{:ru} - Пункт списка
{:de} > Ein Zitat
{:de} | Kopf   | Kopf   |
      | -----| -----|
      | Zelle  | Zelle  |
```

=>

```
<h1 lang="de">Überschrift</h1>
<p lang="fr">Un paragraphe français.</p>
<ul lang="ru">
  <li>Пункт списка</li>
</ul>
<blockquote lang="de">...</blockquote>
<table lang="de">...</table>
```

- **Inline level:** Wrap text with `{:lang}...{:}` to render a `<span lang="...">`:

A French phrase `{:fr}"L'État c'est moi"{:}` is traditionally attributed to King Louis XIV of France.

=>

```
<p>A French phrase <span lang="fr">“L’État c’est moi”</span> is traditionally attributed to King Louis XIV of France</p>
```

## 12. Ruby Annotations

---

```
{日本語|にほんご} => <ruby>日本語<rp>(</rp><rt>にほんご</rt><rp>)</rp></ruby>
```

## 13. Emoji Shortcodes

---

- :joy: 😄
- :smile: 😊
- :heart: ❤️
- :thumbsup: 👍
- :thumbsdown: 👎
- :wink: 😉
- :tada: 🎉
- :rocket: 🚀
- :fire: 🔥
- :star: ★
- :cry: 😭
- :thinking: 🤔
- :100: 💯
- :sparkles: ✨
- :eyes: 👁️
- :bulb: 💡
- :warning: ⚠️
- :ok: 🙆
- :check\_mark: ✓

## 14. Typography and Legal Marks

---

| Name                        | Input   | HTML Output        |
|-----------------------------|---------|--------------------|
| Copyright                   | (c)     | &copy;             |
| Trademark                   | (tm)    | &trade;            |
| Registered Trademark        | (r)     | &reg;              |
| Plus-Minus                  | +/-     | &plusmn;           |
| Not Equal To                | !=      | &ne;               |
| Logical Equivalence         | <=>     | &hArr;             |
| Less-Than or Equal To       | <=      | &le;               |
| Greater-Than or Equal To    | >=      | &ge;               |
| Right Arrow                 | ->      | &rarr;             |
| Left Arrow                  | <-      | &larr;             |
| Up Arrow                    | &uarr;  | &uarr;             |
| Down Arrow                  | &darr;  | &darr;             |
| Logical Implication         | =>      | &rArr;             |
| Solidus (Slash)             | &sol;   | &sol;              |
| Reverse Solidus (Backslash) | &bsol;  | &bsol;             |
| One-Half                    | 1/2     | &frac12;           |
| One-Third                   | 1/3     | &frac13;           |
| Two-Thirds                  | 2/3     | &frac23;           |
| One-Quarter                 | 1/4     | &frac14;           |
| Three-Quarters              | 3/4     | &frac34;           |
| Left Angle Quote            | <<      | &laquo;            |
| Right Angle Quote           | >>      | &raquo;            |
| Left Double Quote           | &ldquo; | &ldquo;            |
| Right Double Quote          | &rdquo; | &rdquo;            |
| Smart Double Quotes         | "text"  | &ldquo;text&rdquo; |
| Smart Single Quotes         | 'text'  | &lsquo;text&rsquo; |
| Straight Apostrophe         | '       | &apos;             |
| Em Dash                     | ---     | &mdash;            |

| Name               | Input  | HTML Output |
|--------------------|--------|-------------|
| En Dash            | --     | &ndash;     |
| Ellipsis           | ...    | &hellip;    |
| Non-Breaking Space | &nbsp; | &nbsp;      |

## 15. Hard Line Breaks

End line with two spaces or backslash: `<br />`

## 16. HTML Comments

[comment]: # => `<!--comment-->`

## 17. YAML Front Matter

If the file begins with a YAML front matter block between `---` lines, the converter emits a complete HTML5 document with the metadata in `<head>` and the body content between `<body>` tags. Without front matter, the output stays a bare HTML fragment. The `--css` option (or `include_css=True`) embeds the default `<style>` block regardless of whether front matter is present.

```
---
lang: en
title: My Document
author: Jane Doe
description: A short description.
keywords: python, markdown, html5
published: 2026-08-09
---
```

- `lang` becomes the `<html lang="...">` attribute (a valid BCP 47 tag).
- `title` becomes the `<title>` element.
- `author`, `description`, `keywords`, and `published` become `<meta name="..." content="..." />` tags.
- A default `<style>` block with viewing-friendly CSS is embedded in `<head>`.

```

<!doctype html>
<html lang="en">
  <head>
    <meta charset="utf-8" />
    <meta name="author" content="Jane Doe" />
    <meta name="description" content="A short description." />
    <meta name="keywords" content="python, markdown, html5" />
    <title>My Document</title>
    <meta name="published" content="2026-08-09" />
    <style>
      /* default viewing-friendly CSS */
    </style>
  </head>
  <body>
    ...
  </body>
</html>

```

Any other keys are ignored, and if the file has no front matter at all, the output remains a bare fragment (unless `--css / include_css=True` is used, in which case it becomes a full document).

## 18. Backslash Escaping

Escape any special char: `\*`, `\#`, `\[`, etc.

Escapable: `\`*_{}[]()#+-.!|~^=:<>`

## 19. Paragraphs

Consecutive text lines merge into `<p>`. Blank lines separate paragraphs.

Empty line after list close => `<!-- -->` preserves whitespace.

## 20. CSS Styles

The converter (with `--css` option or `include_css=True`) embeds a default `<style>` block in `<head>` that provides viewing-friendly styling, regardless of YAML front matter. This predefined CSS rules includes:

```
body {
  padding: 20px;
  font-family: "Noto Serif", "Liberation Serif", "Times New Roman", Times, serif;
  font-size: 18px;
  line-height: 1.4;
  color: #000000;
  background-color: #ffffff;
}
h1 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 32px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
h2 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 28px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
h3 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 24px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
h4 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 20px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
```

```
h5 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 18px;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
h6 {
  margin-top: 1.2em;
  margin-bottom: 0.6em;
  font-family: "Noto Sans", "Liberation Sans", Arial, sans-serif;
  font-weight: bold;
  font-size: 18px;
  font-style: italic;
  hyphens: auto;
  word-break: normal;
  overflow-wrap: break-word;
  text-wrap: balance;
}
p {
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
hr {
  height: 4px;
  margin: 20px 0;
  border: none;
  background-color: #000000;
}
blockquote {
  margin-left: 0;
  padding-left: 20px;
  border-left: 8px solid #f5f5f5;
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
mark {
  padding: 0 2px;
  border-radius: 4px;
  background-color: #ffff00;
  color: #000000;
}
a:link { color: #0000cd; }
a:visited { color: #9400d3; }
a:hover { outline: none; color: #000080; }
```

```
a:focus { outline: none; color: #000080; }
a:active { color: #dc143c; }
ol {
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
ul {
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
li {
  position: relative;
  padding-left: 20px;
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
ul li input[type="checkbox"] {
  margin-right: 4px;
  vertical-align: baseline;
  accent-color: #000000;
  opacity: 1;
  cursor: not-allowed;
}
dt {
  font-weight: bold;
  hyphens: auto;
  word-break: break-word;
  overflow-wrap: anywhere;
}
dd {
  position: relative;
  margin-left: 0;
  padding-left: 20px;
  font-style: italic;
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
code {
  padding: 2px 4px;
  border-radius: 4px;
  font-family: "Noto Sans Mono", "Liberation Mono", "Courier New", Courier, monospace;
  font-size: 0.9em;
  line-height: 1;
  hyphens: none !important;
  white-space: normal;
```

```

word-break: break-all;
overflow-wrap: anywhere;
}
pre {
  max-width: 100%;
  margin: 0;
  padding: 20px;
  border: 1px solid #000000;
  background-color: #f5f5f5;
  overflow: auto;
  scrollbar-color: #000000 transparent;
  white-space: pre-wrap;
  word-break: break-all;
  overflow-wrap: anywhere;
}
pre > code {
  display: block;
  margin: 0;
  padding: 0;
  border: none;
  border-radius: 0;
  line-height: 1.2;
  background-color: transparent;
  overflow: visible;
  hyphens: none !important;
  white-space: pre;
  word-break: normal;
  overflow-wrap: normal;
}
div.code-lang {
  display: block;
  padding: 10px 20px;
  font-family: "Noto Sans Mono", "Liberation Mono", "Courier New", Courier, monospace;
  font-size: 0.9em;
  line-height: 1;
  background-color: #000000;
  color: #ffffff;
  font-weight: bold;
}
table {
  margin: 20px 0;
  border-collapse: collapse;
}
th { padding: 10px 12px; border: 1px solid #000000; font-weight: bold; }
td { padding: 10px 12px; border: 1px solid #000000; }
thead tr { background-color: #000000; color: #ffffff; }
thead th { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }
thead td { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }
tbody th { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }
tbody td { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }
tfoot tr { background-color: #f5f5f5; font-style: italic; }
tfoot th { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }
tfoot td { hyphens: auto; word-break: break-word; overflow-wrap: anywhere; }

```

```

figure {
  display: block;
  margin: 0;
}
figure img {
  display: block;
  max-width: 100%;
  height: auto;
}
figcaption {
  text-align: left;
  font-style: italic;
  hyphens: auto;
  hyphenate-limit-chars: 6 3 3;
  word-break: normal;
  overflow-wrap: break-word;
}
ruby { ruby-position: over; ruby-align: space-around; }
rt {
  letter-spacing: 0.05em;
  font-size: 0.55em;
  line-break: strict;
  white-space: nowrap;
  overflow-wrap: normal;
}
rp { display: none; }
span[lang="ja"] {
  font-family: "Noto Serif CJK JP", "Source Han Serif JP", "源ノ明朝", "Source Han Serif", "Hiragino Mincho ProN", "Hiragino Mincho Pro", "IPAexMincho", "IPAMincho", "MS PMincho", "MS Mincho", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-CN"] {
  font-family: "Noto Serif CJK SC", "Source Han Serif SC", "思源宋体", "Source Han Serif CN", "Source Han Serif", "Songti SC", "FandolSong", "WenQuanYi Bitmap Song", "SimSun", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-Hans"] {
  font-family: "Noto Serif CJK SC", "Source Han Serif SC", "思源宋体", "Source Han Serif CN", "Source Han Serif", "Songti SC", "FandolSong", "WenQuanYi Bitmap Song", "SimSun", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-TW"] {
  font-family: "Noto Serif CJK TC", "Source Han Serif TC", "思源宋體", "Source Han Serif TW", "Source Han Serif", "Apple LiSung", "LiSong Pro", "HanaMinA", "PMingLiU", "MingLiU", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="zh-Hant"] {
  font-family: "Noto Serif CJK TC", "Source Han Serif TC", "思源宋體", "Source Han Serif TW", "Source Han Serif", "Apple LiSung", "LiSong Pro", "HanaMinA", "PMingLiU", "MingLiU", serif;

```

```
word-break: break-all;
line-break: normal;
}
span[lang="zh-HK"] {
  font-family: "Noto Serif CJK HK", "Source Han Serif HK", "思源宋體 香港", "思源宋體", "Source Han
Serif", "Apple LiSung", "LiSong Pro", "HanaMinA", "MingLiU_HKSCS", "PMingLiU", "MingLiU", serif;
  word-break: break-all;
  line-break: normal;
}
span[lang="ko"] {
  font-family: "Noto Serif CJK KR", "Source Han Serif KR", "본명조", "Source Han Serif",
"AppleMyungjo", "UnBatang", "은바탕", "Batang", serif;
  word-break: break-all;
  line-break: normal;
}
```